

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Perihal Produk: **2,2,2-Trifluoroethanol (Peptide Synthesis)**  
 Product Description: **2,2,2-Trifluoroethanol (Peptide Synthesis)**  
 Cat No. : BP622-100  
 Synonyms TFE  
 CAS No 75-89-8  
 Molecular Formula C<sub>2</sub> H<sub>3</sub> F<sub>3</sub> O

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use Laboratory chemicals.  
 Uses advised against No Information available

**Company**

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 CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                                                      |                     |
|------------------------------------------------------|---------------------|
| Flammable liquids                                    | Category 3 (H226)   |
| Acute oral toxicity                                  | Category 3 (H301)   |
| Acute Inhalation Toxicity - Vapors                   | Category 3 (H331)   |
| Serious Eye Damage/Eye Irritation                    | Category 1 (H318)   |
| Reproductive Toxicity                                | Category 1B (H360F) |
| Specific target organ toxicity - (repeated exposure) | Category 2 (H373)   |

**Label Elements**


Signal Word

Danger

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## Hazard Statements

H226 - Flammable liquid and vapor  
H318 - Causes serious eye damage  
H360F - May damage fertility  
H373 - May cause damage to organs through prolonged or repeated exposure  
H301 + H331 - Toxic if swallowed or if inhaled

## Precautionary Statements

### Prevention

P201 - Obtain special instructions before use  
P202 - Do not handle until all safety precautions have been read and understood  
P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P240 - Ground and bond container and receiving equipment  
P241 - Use explosion-proof electrical/ ventilating/ lighting equipment  
P242 - Use non-sparking tools  
P243 - Take action to prevent static discharges  
P260 - Do not breathe dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

### Response

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor  
P330 - Rinse mouth  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

### Storage

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed  
P405 - Store locked up

### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Other Hazards

Toxic to terrestrial vertebrates  
This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component              | CAS No  | Weight % |
|------------------------|---------|----------|
| 2,2,2-Trifluoroethanol | 75-89-8 | <=100    |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical

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advice.

## Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

## Ingestion

Do NOT induce vomiting. Call a physician or poison control center immediately.

## Inhalation

Remove to fresh air. If breathing is difficult, give oxygen. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.

## Self-Protection of the First Aider

Use personal protective equipment as required.

## Most important symptoms and effects, both acute and delayed

Difficulty in breathing. Causes eye burns. . Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting.

## Indication of any immediate medical attention and special treatment needed

### Notes to Physician

Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### Suitable Extinguishing Media

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers.

#### Extinguishing media which must not be used for safety reasons

No information available.

### Special hazards arising from the substance or mixture

Flammable. Risk of ignition. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Containers may explode when heated. Keep product and empty container away from heat and sources of ignition. Vapors may form explosive mixtures with air.

### Hazardous Combustion Products

Thermal decomposition can lead to release of irritating gases and vapors, Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Gaseous hydrogen fluoride (HF).

### Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Use personal protective equipment as required. Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas. Remove all sources of ignition. Take precautionary measures against static discharges.

### Environmental precautions

Should not be released into the environment. See Section 12 for additional Ecological Information.

### Methods and Material for Containment and Cleaning Up

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Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

## Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Take precautionary measures against static discharges.

### Conditions for Safe Storage, Including any Incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

### Exposure Controls

#### Engineering Measures

Use only under a chemical fume hood. Use explosion-proof electrical/ventilating/lighting equipment. Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles

#### Hand Protection

Protective gloves

#### Skin and body protection

Long sleeved clothing

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

#### Recommended Filter type:

Organic gases and vapours filter Type A Brown conforming to EN14387

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

When RPE is used a face piece Fit Test should be conducted

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## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

## Environmental exposure controls

No information available

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

|                                         |                             |                                        |
|-----------------------------------------|-----------------------------|----------------------------------------|
| Appearance                              | Colorless                   |                                        |
| Physical State                          | Liquid                      |                                        |
| Odor                                    | Characteristic              |                                        |
| Odor Threshold                          | No data available           |                                        |
| pH                                      | 5.0-7.5                     |                                        |
| Melting Point/Range                     | -45 °C / -49 °F             |                                        |
| Softening Point                         | No data available           |                                        |
| Boiling Point/Range                     | 77 - 80 °C / 170.6 - 176 °F |                                        |
| Flash Point                             | 29 °C / 84.2 °F             | Method - No information available      |
| Evaporation Rate                        | No data available           |                                        |
| Flammability (solid,gas)                | Not applicable              | Liquid                                 |
| Explosion Limits                        | Lower 5.5<br>Upper 42       |                                        |
| Vapor Pressure                          | 70 mmHg @ 25°C              |                                        |
| Vapor Density                           | 3.45                        | (Air = 1.0)                            |
| Specific Gravity / Density              | 1.390                       |                                        |
| Bulk Density                            | Not applicable              | Liquid                                 |
| Water Solubility                        | Soluble                     |                                        |
| Solubility in other solvents            | No information available    |                                        |
| Partition Coefficient (n-octanol/water) |                             |                                        |
| Component                               | log Pow                     |                                        |
| 2,2,2-Trifluoroethanol                  | <0.3                        |                                        |
| Autoignition Temperature                | 480 - °C / 896 - °F         |                                        |
| Decomposition Temperature               | No data available           |                                        |
| Viscosity                               | 1.75 mPa.s @ 25 °C          |                                        |
| Explosive Properties                    |                             | explosive air/vapour mixtures possible |
| Oxidizing Properties                    | No information available    |                                        |
| Molecular Formula                       | C2 H3 F3 O                  |                                        |
| Molecular Weight                        | 100.04                      |                                        |

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

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## Chemical Stability

Stable under normal conditions.

## Possibility of Hazardous Reactions

**Hazardous Polymerization**  
**Hazardous Reactions**

No information available.  
None under normal processing.

## Conditions to Avoid

Incompatible products. Excess heat. Keep away from open flames, hot surfaces and sources of ignition.

## Incompatible Materials

Bases. Metals. Strong oxidizing agents.

## Hazardous Decomposition Products

Thermal decomposition can lead to release of irritating gases and vapors. Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Gaseous hydrogen fluoride (HF).

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### Product Information

##### (a) acute toxicity;

**Oral**

Category 3

**Dermal**

Based on available data, the classification criteria are not met

**Inhalation**

Category 3

| Component              | LD50 Oral                    | LD50 Dermal              | LC50 Inhalation              |
|------------------------|------------------------------|--------------------------|------------------------------|
| 2,2,2-Trifluoroethanol | LD50 153 - 177 mg/kg ( Rat ) | LD50 >2000 mg/kg ( Rat ) | LC50 = 3,25 mg/L ( Rat ) 4 h |

##### (b) skin corrosion/irritation;

Based on available data, the classification criteria are not met

##### (c) serious eye damage/irritation;

Category 1

##### (d) respiratory or skin sensitization;

**Respiratory**

Based on available data, the classification criteria are not met

**Skin**

Based on available data, the classification criteria are not met

##### (e) germ cell mutagenicity;

Based on available data, the classification criteria are not met

Not mutagenic in AMES Test

##### (f) carcinogenicity;

Based on available data, the classification criteria are not met

There are no known carcinogenic chemicals in this product

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|                                                    |                                                                                                                                     |
|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| (g) reproductive toxicity;<br>Reproductive Effects | Category 1B<br>May impair fertility.                                                                                                |
| (h) STOT-single exposure;                          | Based on available data, the classification criteria are not met                                                                    |
| (i) STOT-repeated exposure;<br>Target Organs       | Category 2<br>Blood.                                                                                                                |
| (j) aspiration hazard;                             | Based on available data, the classification criteria are not met                                                                    |
| Other Adverse Effects                              | The toxicological properties have not been fully investigated. See actual entry in RTECS for complete information                   |
| Symptoms / effects, both acute and delayed         | Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting.                                                |
| Endocrine Disrupting Properties                    | Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors. |

## SECTION 12: ECOLOGICAL INFORMATION

Ecotoxicity effects Do not empty into drains.

| Component              | Freshwater Fish                                                    | Water Flea | Freshwater Algae | Microtox |
|------------------------|--------------------------------------------------------------------|------------|------------------|----------|
| 2,2,2-Trifluoroethanol | LC50: 105 - 135 mg/L,<br>96h flow-through<br>(Pimephales promelas) |            |                  |          |

Persistence and degradability  
**Persistence** Not readily biodegradable  
Persistence is unlikely, based on information available.

Bioaccumulative potential Bioaccumulation is unlikely

| Component              | log Pow | Bioconcentration factor (BCF) |
|------------------------|---------|-------------------------------|
| 2,2,2-Trifluoroethanol | <0.3    | No data available             |

Mobility in soil The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces. Will likely be mobile in the environment due to its volatility. Disperses rapidly in air.

Endocrine Disruptor Information This product does not contain any known or suspected endocrine disruptors

Other adverse effects No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

Waste treatment methods  
**Waste from Residues/Unused Products** Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations

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**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous Keep product and empty container away from heat and sources of ignition

**Other Information** Waste codes should be assigned by the user based on the application for which the product was used Do not flush to sewer Can be landfilled or incinerated, when in compliance with local regulations Do not empty into drains

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

UN-No UN1992  
Hazard Class 3  
Subsidiary Hazard Class 6.1  
Packing Group III  
Proper Shipping Name Flammable liquid, toxic, n.o.s. 2,2,2-Trifluoroethanol

### Road and Rail Transport

UN-No UN1992  
Hazard Class 3  
Subsidiary Hazard Class 6.1  
Packing Group III  
Proper Shipping Name Flammable liquid, toxic, n.o.s. 2,2,2-Trifluoroethanol

### IATA

UN-No UN1992  
Hazard Class 3  
Subsidiary Hazard Class 6.1  
Packing Group III  
Proper Shipping Name Flammable liquid, toxic, n.o.s. 2,2,2-Trifluoroethanol

**Special Precautions for User** No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

**International Inventories** X = listed

| Component              | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL |
|------------------------|-----------|------|-----|-------|------|------|-------|------|------|
| 2,2,2-Trifluoroethanol | 200-913-6 | X    | X   | X     | X    | X    | X     | X    | -    |

### National Regulations

**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION



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## Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

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23-Mar-2025

Revision Summary

Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**